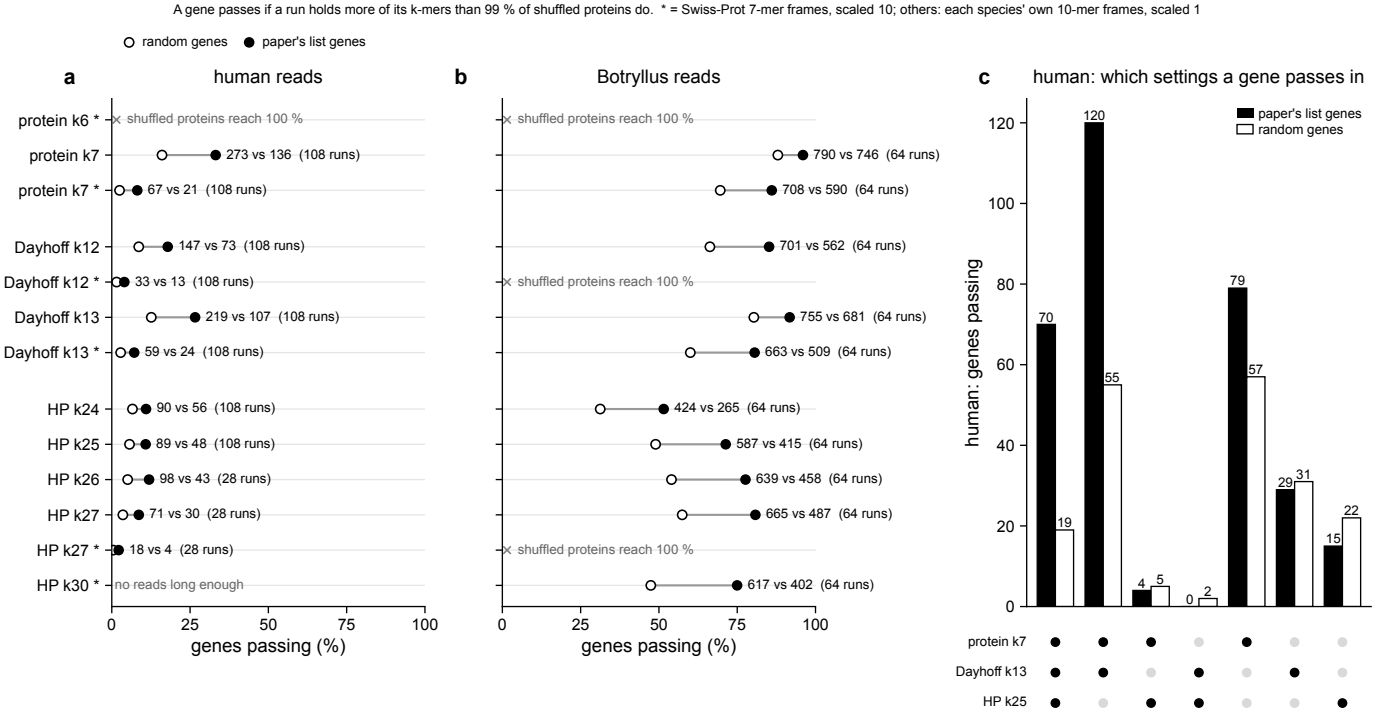
A gene passes if a run holds more of its k-mers than 99 % of shuffled proteins do. * = Swiss-Prot 7-mer frames, scaled 10; others: each species' own 10-mer frames, scaled 1
random genes
paper's list genes
a
human reads
protein k6 *
shuffled proteins reach 100 %
protein k7
273 vs 136 (108 runs)
protein k7 *
67 vs 21 (108 runs)
Dayhoff k12
147 vs 73 (108 runs)
Dayhoff k12 *
33 vs 13 (108 runs)
Dayhoff k13
219 vs 107 (108 runs)
Dayhoff k13 *
59 vs 24 (108 runs)
HP k24
90 vs 56 (108 runs)
HP k25
89 vs 48 (108 runs)
HP k26
98 vs 43 (28 runs)
HP k27
71 vs 30 (28 runs)
HP k27 *
18 vs 4 (28 runs)
HP k30 *
no reads long enough
0
25
50
75
100
genes passing (%)
b
Botryllus reads
shuffled proteins reach 100 %
790 vs 746 (64 runs)
708 vs 590 (64 runs)
701 vs 562 (64 runs)
shuffled proteins reach 100 %
755 vs 681 (64 runs)
663 vs 509 (64 runs)
424 vs 265 (64 runs)
587 vs 415 (64 runs)
639 vs 458 (64 runs)
665 vs 487 (64 runs)
shuffled proteins reach 100 %
617 vs 402 (64 runs)
0
25
50
75
100
genes passing (%)
c
human: which settings a gene passes in
paper's list genes
random genes
human: genes passing
0
20
40
60
80
100
120
70
19
120
55
4
5
0
2
79
57
29
31
15
22
protein k7
Dayhoff k13
HP k25